

Web browsers are useless here.

```
,+++777777++=:,  
7~?7 +7I77 :,I777 I 77 7  
=7I7I~7 ,77: ++:~+7 77=7777 7 +77=7  
77+7I 777~, ,=7~ ,::7=7: 7 77 77: 7 7  
= 7 ?7 , 7~,~ + 77 ?: :?777 +~77 77?  
77 ~I == ~77= +777 777~: I,+77? 7  
I 7=77~+77+?= :I+~77? , I 7? 7  
=77 77= +7 7777 ,7 7??:  
=I, I 7+:77? +7I7?7  
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I777=7777 ~ 77 :  
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77:I  
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```

Welcome.

Good luck.

>_

Cracking Cicada 3301

The Future of Collaborative Puzzle-Solving

A CicadaSolvers Production
By Taiwo, Artorias, Puck, and Clockwork

What is Cicada 3301 and Who are we?

Cicada 3301

- Mysterious international organization that released a series of increasingly difficult cryptographic puzzles.
- The current step of the 2014 puzzle, a digital book of runes called Liber Primus, has gone unsolved.

CicadaSolvers

The main community of people interested in the Cicada 3301 puzzles.

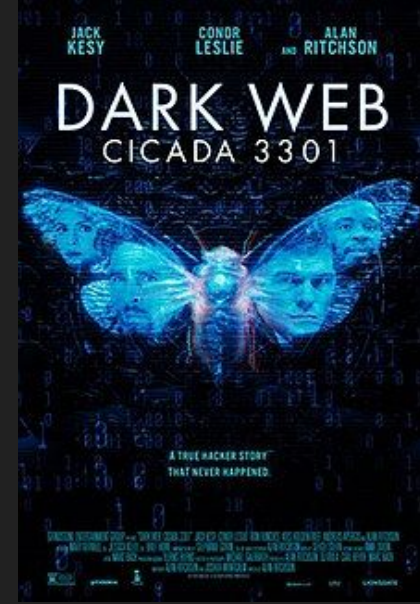
Speakers

- Puck - Community Organizer
- Taiwo - Technologist
- Artorias - Archivist
- Clockwork - Server Engagement

Misrepresentation in Popular Media

Inaccurate portrayals of Cicada 3301 in popular media

- Youtube videos
- Social Media
- Fake Puzzles
- News articles
- Movie



The Reality: Who is Cicada 3301?

What we have observed:

- Refer to themselves as 3301 with a plural self-reference
- They claim to be recruiting intelligent individuals
- Have the resources to construct complex puzzles with global reach
- Advocates of privacy and skilled with anonymity
- Verify all messages with PGP 7A35090F
- References to esoterica are a thematic choice

Common baseless claims:

- Criminal organisation
- Government agency
- Advertising campaign
- Cult / Religious origin
- Liber Primus is solved
- Puzzle releases since 2014

-----BEGIN PGP SIGNED MESSAGE-----

Hash: SHA1

DO NOT SHARE THIS INFORMATION!

Congratulations. Your testing has finally come to an end. We hope you have enjoyed the "vacation" over the last few weeks. You will be very busy now should you choose to join us.

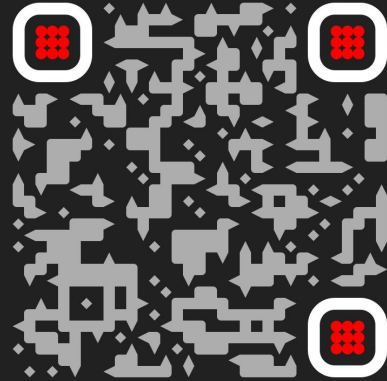
There are two final steps, although there won't be any hidden codes, or secret messages, or physical treasure hunts. This first of these is only honesty. We have always been honest with you, and we shall continue to be honest with you. And we expect you to be honest with us in return.

The Leaked Email

You have all wondered who we are, and so we shall now tell you : **We are an international group.** We have no name. We have no symbol. We have no membership rosters. **We do not have a public website, and we do not advertise ourselves.** We are a group of individuals who have proven ourselves. Much of our time is spent in planning this recruitment contest. And we are drawn together by common beliefs. A careful reading of the texts used in the contest would have revealed some of our beliefs : that tyranny and oppression of any kind must end; that censorship is wrong and that privacy is an inalienable right.

We are not a "hacker" group. Nor are we a "warez" group. **We do not engage in illegal activity,** nor do our members. If you are engaged in illegal activity, we ask that you cease any and all illegal activities or decline membership at this time. We will not ask questions if you decline; however, if you lie to us we will find out.

You are undoubtedly wondering what it is that we do. **We are much like a "Think Tank"** in that our primary focus is on researching and developing techniques to aid the ideas we advocate : **liberty, privacy, security.** You have undoubtedly heard of a few of our past projects. And if you choose to accept membership, we are happy to have you on board to help with future projects.



Read it here

We look forward to hearing from you.

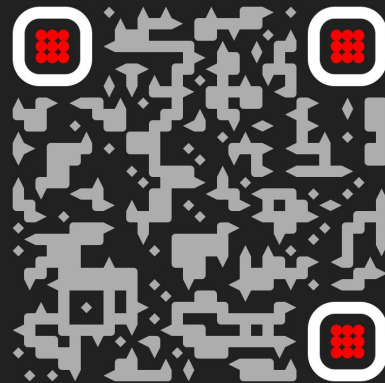
3301

-----BEGIN PGP SIGNATURE-----
Version: GnuPG v1.4.11 (GNU/Linux)

iQIcBAEBAgAGBQJURhuCKAAoJEBgfAeV6N0KpJCoQALZk/jfmJ0bu0xu4jI++cR77
hJIE9yOhTgNebcKPNYv5xBbJJEKrvd3dZ2YhK067227Zf2Y+1LkonRtkqyFFHRX
mKASFYFmg7kyL5+kNxe5EZvgECAC21ADhmI8cFxsWgnWEXO/KE++raJDJ3jBzW5
hwscm0Bg/KpsEZTONwLm0d0j6RjKrBzVeQ11xdaQXD9Np4/t3+SwoAjlVqVezK
9eICCa10cydnhr+lgBL0TgAw7qHfLMIISxahAkpCoQ0S8h/rph9PpAg/cumy4
Shoem01WtVt23ysFnd+vF9pYxatun3pYb0pmmrkeJ0S3ze/1ISEQ/0W/SP
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Yv9ChzVd6z3pRRHAgkf210sadrS5qx1jaBEV64J/LJL8KzF4t0Vx04rV4FkrN
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vJB8iPO8eCsu0Q0bn6rr
=e31t
-----END PGP SIGNATURE-----

”

Key Points From The Leaked Email



[Read it here](#)

- “We are an **international** group.”
- “We are drawn together by common beliefs ... **censorship** is wrong and ... **privacy** is an inalienable right.”
- “We are much like a **think tank**”
- “The ideas we advocate: **Liberty, Privacy, Security**”

Similarities with Cypherpunk Philosophies

Cypherpunk Mindset - “Privacy is necessary for an open society in the electronic age...freedom of speech, even more than privacy, is fundamental to an open society...Privacy in an open society also requires cryptography...Cypherpunks deplore regulations on cryptography, for encryption is fundamentally a private act...” - **Eric Hughes Cypherpunk Manifesto**

“For privacy to be widespread it must be part of a social contract. People must come and together deploy these systems for the common good. Privacy only extends so far as the cooperation of one's fellows in society.”

- 3301's puzzles actively extend the range of people deploying privacy ensuring systems

The Puzzle Timeline

Gematria Primus

an order and a value as revealed through 3301

Rune	Letter	Value
𐌲	AE	101
𐌸	Y	103
𐌶	IA/IO	107
𐌹	EA	109

The Grand Entrance

- PGP Signed messages
- Hidden services
- Phone numbers
- Cracking RSA keys
- Global Posters

The Publication of The Liber Primus

- 17 pages of solved runes
- Cracking RSA keys (again)
- Interconnectedness
- The 58 page 'LP2'
- The deep web hash

2013

Now

2012

2014

Round Two

- Instar Emergence
- Gematria Primus
- CicadaOS
- Web console
- Global posters (again)

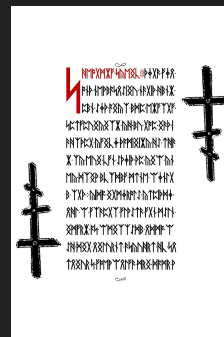
Hello. We are looking for highly intelligent individuals. To find them, we have devised a test.

There is a message hidden in this image.

Find it, and it will lead you on the road to finding us. We look forward to meeting the few that will make it all the way through.

Good luck.

3301

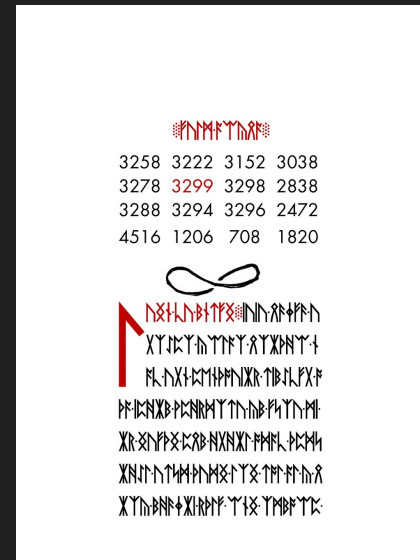
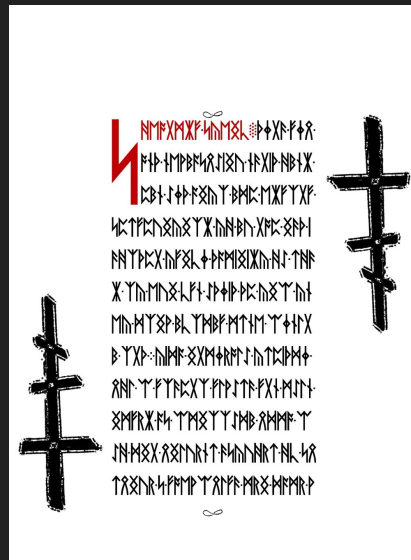


The Unsolved Step: Liber Primus

In May 2014, the latest step was released: A 58-page digital book of elder futhark runes.

Two pages were solved immediately; the rest have resisted thousands of attempts.

- Pages are uniformly sized JPG files
- Some pages are greyscale
- Runes are encrypted
- Normal statistical distribution
- Repeating bi-grams have a high frequency





36367763ab73783c7af284446c
59466b4cd653239a311cb7116
d4618dee09a8425893dc7500b
464fdaf1672d7bef5e891c6e227
4568926a49fb4f45132c2a8b4

LMFFXMFILXRFNRMMHNTN
XPYFMYPFRXXXRBHNN



Page 56 - The Deep Web Hash

AN END - WITHIN THE **DEEP WEB**, THERE
EXISTS A PAGE THAT HASHES TO:

- 512 bit hash
- 128 character hex
- Relatively low value

IT IS THE DUTY OF EVERY PILGRIM TO SEEK
OUT **THIS PAGE**.

The CicadaSolvers Community

The path lies empty, but we are undeterred. Tens of thousands of people are still working to decrypt the runes and solve the internet's hardest mystery.

The Discord community alone has well over 7,000 members, numerous ongoing solving attempts, and regular community events.

Uncovering Cicada

wikia



CicadaSolvers

● 1,026 Online ● 7,338 Members



Community Evolution

4Chan

- The origin of the puzzle
- Unpredictable community (toxic)
- Liberated by the lack of moderation and accountability

IRC

- Pre-existing community of ARG enthusiasts
- Low accessibility means lower user count but higher average ability
- Accessible privacy and open sourced clients
- Low capability to moderate and organise content

Discord

- Proprietary platform run by data miners
- Huge user-base
- Large set of moderation and organisational features
- Strict terms of service

CicadaSolvers - What makes us different

Crowdsourced Information

Dedication to the sharing of progress and tools.

Stable Definition of Validity

No PGP = No Cicada.

Democratic Moderation

Clear rules, separation of personal interactions, and sanctions.

Inclusive Solving Atmosphere

Guide and teach rather than criticize and ridicule.

Structured Community Effort

Solving sessions, documentation processes, collaborative efforts, and community resources.

Our Main Challenges

Documentation

There are thousands of solving attempts every year, but very few are shared publicly. Furthermore, cryptanalyses of the current step are hard to reach.



- Centralize analyses
- Make documentation easier
- Promote documentation

Fake Puzzles

Fake puzzles are pervasive in the community. People commonly create puzzles that utilizes 3301's fame to target inexperienced solvers.



- Flag fake puzzles without popularization
- Enlist lost solvers
- Avoid fake puzzles

Revolving Door

Solvers come and go all the time, so the active solving community is constantly adapting to new members and filling the shoes of old solvers.



- Increase solver numbers
- Maintain members

The Documentation Challenge

1. Where do we find analyses of Liber Primus?

- a. Resources in Discord server
- b. Uncovering Cicada Wikia
- c. Individual solvers' efforts to centralize

2. How do we find past solving attempts?

- a. Discord search feature
 - i. Specific channels for specific topics
- b. Potentially building systems to handle attempts
 - i. WeKan / Trello
 - ii. Forums
 - iii. "Solving cicada" github

The Revolving Door



Our Main Advantages

Large Community

With thousands of members in our main Discord server, we always have multiple ongoing solving attempts.



- How does community size correlate with member retention?

Organization of Manpower

With such a large community, we have developed strategies to optimize collaborative solving.



- How do our community events work?
- How do we share workload?

Active Tool Development

Solvers have streamlined the solving process through the creation of tools.



- What tools do we have?
- How do we make tools easier to build?

Benefits of Our Community's Size

- Constant flow of new attempts and people eager to learn
- The more solvers there are, the easier it is for new solvers to stick around
 - Older solvers can help out newer solvers
 - More motivation to solve when there are other people
 - More solvers → more solving ideas → you are more likely to find an idea you like → more solving motivation

Organization of Manpower

Voice Sessions

- Weekly and monthly events since January 2020 where community members gather in a voice channel
- Best place to share ideas and solve together

Teaming up on solving

- Voice sessions allow for sharing of workload
- Delineation of tasks
- Vet ideas

The Creation and Usage of Tools

Tools we have

- Cyberchef adaptation - General tool for ciphers adapted to support runes
- TaiiwoBot (CicadaBot) - Discord and IRC bot for rapid prototyping and more
- Many specialized scripts to run tests and ciphers
- 3301 Hash alarm - hashes browser pages automatically
- Many more - Cataloguing these tools is a project of it's own

Making tool development easier

- Taiiwo's cicada library - OOP Python library for rune conversion and more
- Open source culture

The Future Will Prevail

This year's DEF CON theme is 'The Future Will Prevail'

- The CicadaSolvers community embodies this spirit
- We have worked for years against all odds to solve the the unsolvable
- If we don't stay optimistic, we know we will give up
- The future will prevail because people are learning now



An End

Recap

- Overview of Cicada 3301
- Puzzle disinformation
- The Liber Primus and the deep web hash
- The CicadaSolvers community
- Our approach to collaborative puzzle-solving
- The future will prevail

Important Links

- Uncovering Cicada Wikia
- cicada.gq/discord
- github.com/cicada-solvers